

# A Bound-Domain Vacuum Response Model for SPARC Flat Rotation Velocities

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# Abstract

We present a phenomenological galaxy-sector response law motivated by the Conformal Gravity Hilbert Space Time Current framework. The model, called BDVR, interprets the dark-matter-like rotation signal as an effective bound-domain response of coherent galactic baryonic systems rather than as a freely tuned per-galaxy halo. The frozen implementation tested here combines a radial halo-shaped response envelope with observable baryonic-structure conditioning through stellar/gas composition and morphology. Using a SPARC quality-1 flat-velocity sample, we evaluate the frozen M4c response with no per-galaxy  $Q$  tuning, no retuning on the validation sample, and no outlier deletion. The frozen M4c model reaches a median absolute  $\log_{10}$  velocity residual of 0.0416 dex on 87 galaxies, with 83 of 87 galaxies within 0.10 dex and 86 of 87 within 0.15 dex. Baseline and ablation comparisons show that radial halo activation and baryonic-structure conditioning improve the residual behavior relative to simpler forms. The leading residual object, UGC07125, is retained and treated as an external verification target for kinematic integrity, gas participation, and HI morphology rather than tuned away. We frame BDVR as a falsifiable galaxy-sector effective law, not as a complete replacement for dark matter or a first-principles derivation.

## I. INTRODUCTION

Galaxy rotation curves remain one of the central empirical motivations for dark matter or modified gravitational dynamics. A successful phenomenological model must reproduce flat rotation behavior while avoiding excessive galaxy-by-galaxy freedom. The present work introduces BDVR as a galaxy-sector effective law motivated by the Conformal Gravity Hilbert Space Time Current framework.

The central idea is that bound coherent structures can couple to an underlying vacuum/time-current reservoir, while extended null propagation can remain nearly propagation-safe. In this paper we restrict attention to the galaxy sector and test whether a globally parameterized response law can reproduce SPARC flat-velocity behavior.

We do not claim a complete action-level derivation. The objective is narrower: define the effective response law, freeze its parameters, test it on a quality-controlled sample, compare it with baselines and ablations, and identify retained failure modes.

## II. CONFORMAL GRAVITY HILBERT SPACE TIME CURRENT SECTOR PICTURE

The parent framework distinguishes observational sectors. Propagation observables such as supernova, BAO, and CMB distances are treated as extended null or decohered paths that should remain nearly LCDM-safe. Local bound clocks and galaxies, by contrast, are treated as coherent bound domains that may couple to the time-current reservoir.

The sector hierarchy is:

$$\text{Conformal Gravity Hilbert Space Time Current} \rightarrow \text{propagation sector}, \quad (1)$$

$$\rightarrow \text{calibration sector}, \quad (2)$$

$$\rightarrow \text{galaxy BDVR sector}. \quad (3)$$

In this paper, BDVR is the galaxy-sector realization of this idea.

## III. BDVR EFFECTIVE LAW

The radial diagnostic acceleration form is

$$a_{\text{obs}}(r) = a_b(r) + a_{\text{BDVR}}(r), \quad (4)$$

with

$$a_{\text{BDVR}}(r) = Q_{\text{eff}}(r, g) \sqrt{a_b(r) a_0}. \quad (5)$$

The galaxy-sector response factor is

$$Q_{\text{eff}}(r, g) = Q_0 H(r/R_d) S_{\text{sg}}(g) S_{\text{int}}(g). \quad (6)$$

The radial halo activation is

$$H(r/R_d) = 1 + A_h \left[ 1 - \exp \left( -\frac{r}{k_h R_d} \right) \right]. \quad (7)$$

The stellar/gas structural factor is

$$S_{\text{sg}} = 1 + A_{\text{sg}} \max \left[ 0, \log_{10} \left( \frac{M_*}{M_g} \right) - \log_{10}(5) \right]. \quad (8)$$

The morphology factor is

$$S_{\text{int}} = 1 + A_{\text{int}} \max(0, 5 - T). \quad (9)$$

TABLE I. Frozen parameters used in the M4c/M5 BDVR validation. All values are global and are not adjusted per galaxy.

| Parameter            | Value              | Description                                            |
|----------------------|--------------------|--------------------------------------------------------|
| q_base               | 0.9002495108803148 | Global baseline BDVR galaxy-sector response amplitude  |
| a0_m_s2              | 1.2e-10            | Reference acceleration scale                           |
| halo_amplitude       | 0.35               | Amplitude of radial halo activation                    |
| halo_k               | 3.0                | Halo scale length in units of Rdisk                    |
| r_eval_mode          | transition         | Adaptive radius mode used to evaluate $Q_{\text{eff}}$ |
| velocity_mode        | flat_response      | Flat-response estimator for SPARC Table-1 Vflat        |
| r_eval_multiple      | 4.0                | Fallback r_eval/Rdisk                                  |
| min_r_eval_multiple  | 1.5                | Minimum clipped r_eval/Rdisk                           |
| max_r_eval_multiple  | 12.0               | Maximum clipped r_eval/Rdisk                           |
| structural_mode      | star_gas_hinge     | Star/gas structural response mode                      |
| structural_amplitude | 0.3                | Star/gas hinge amplitude                               |
| star_gas_threshold   | 5.0                | Threshold for stellar-to-gas ratio hinge               |
| internal_mode        | early_type_hinge   | Internal morphology response mode                      |
| internal_amplitude   | 0.1                | Early-type morphology hinge amplitude                  |
| early_type_threshold | 5.0                | Morphological T threshold for early-type response      |

For SPARC Table-1 flat-velocity comparisons we use

$$v_{\text{flat}}^{\text{BDVR}} = (Q_{\text{eff}}^2 G M_b a_0)^{1/4}. \quad (10)$$

The adaptive radius enters only through  $Q_{\text{eff}}$ . The SPARC flat-velocity comparison is not computed as  $\sqrt{r a_{\text{total}}}$ .

#### IV. FROZEN PARAMETER SET

#### V. DATA AND VALIDATION PROTOCOL

The validation uses a SPARC quality-1 sample with positive  $V_{\text{flat}}$ ,  $L_{3.6}$ ,  $M_{\text{HI}}$ , and  $R_d$ . The frozen model is evaluated with no per-galaxy  $Q$  adjustment, no retuning on the validation sample, and no deletion of the leading outlier.

TABLE II. Frozen M4c validation summary for the SPARC quality-1 flat-velocity sample.

| Metric                              | Value                | Description                                      |
|-------------------------------------|----------------------|--------------------------------------------------|
| n_galaxies                          | 87                   | SPARC quality-1 galaxies in frozen M5 validation |
| median_abs_log10_velocity_error     | 0.041613899469354995 | Median absolute log10(pred/obs)                  |
| mean_abs_log10_velocity_error       | 0.04572143797277067  | Mean absolute log10(pred/obs)                    |
| max_abs_log10_velocity_error        | 0.23214012979801618  | Maximum absolute log10(pred/obs)                 |
| median_velocity_ratio_pred_over_obs | 1.0215021046046753   | Median predicted/observed velocity ratio         |
| frac_within_010_dex                 | 0.9540229885057471   | Fraction within 0.10 dex                         |
| frac_within_015_dex                 | 0.9885057471264368   | Fraction within 0.15 dex                         |
| frac_within_020_dex                 | 0.9885057471264368   | Fraction within 0.20 dex                         |
| n_within_010_dex                    | 83                   | Count within 0.10 dex                            |
| n_within_015_dex                    | 86                   | Count within 0.15 dex                            |
| n_within_020_dex                    | 86                   | Count within 0.20 dex                            |
| n_above_030_dex                     | 0                    | Count above 0.30 dex                             |
| strong_pass                         | False                | Draft strong-pass gate                           |
| soft_pass                           | True                 | Draft soft-pass gate                             |
| fail                                | False                | Draft fail gate                                  |

The stellar and gas masses are constructed as

$$M_* = 0.5L_{3.6} \times 10^9 M_\odot, \quad (11)$$

and

$$M_g = 1.33M_{\text{HI}} \times 10^9 M_\odot. \quad (12)$$

## VI. FROZEN M4C VALIDATION RESULT

The model gives a median absolute log<sub>10</sub> residual of 0.0416 dex, a mean absolute residual of 0.0457 dex, and a maximum absolute residual of 0.2321 dex. The fraction within 0.10 dex is 0.9540; the fraction within 0.15 dex is 0.9885. No galaxy exceeds 0.30 dex.

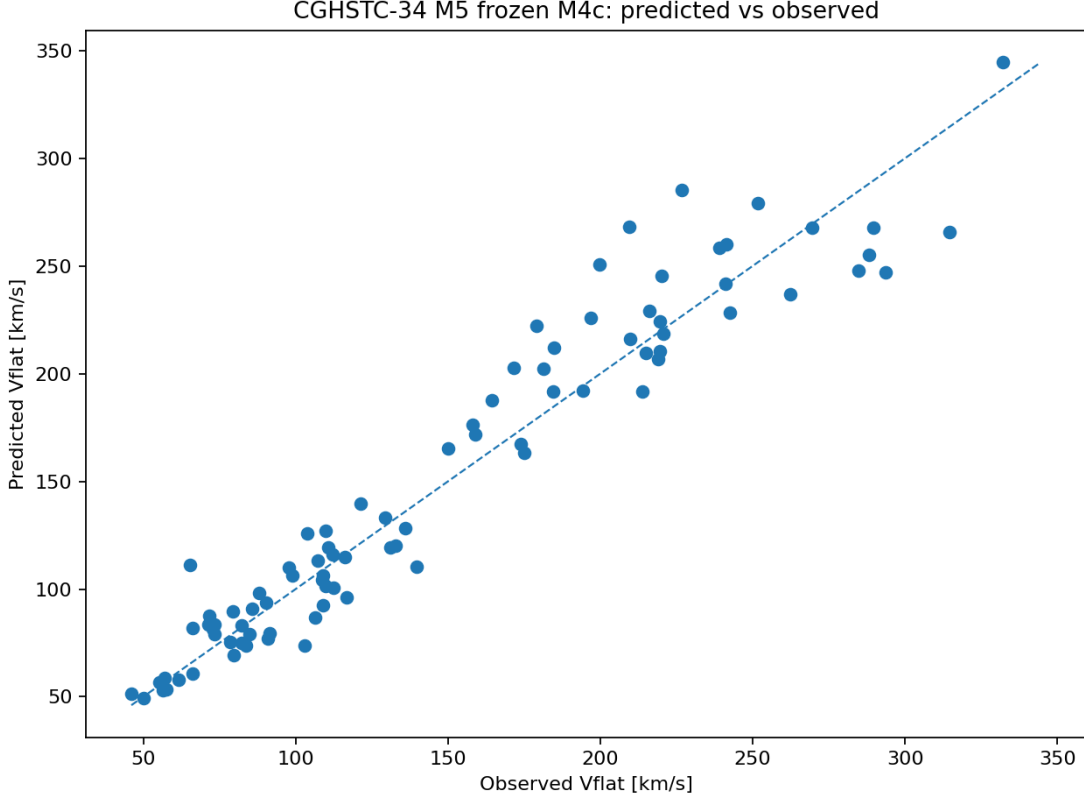


FIG. 1. Frozen M4c BDVR predicted versus observed SPARC flat velocities for the quality-1 validation sample.

## VII. BASELINE AND ABLATION COMPARISON

The baseline comparison evaluates Newtonian baryons, a simple BTFR/MOND-like estimator, scalar- $Q$ , halo-only, halo plus star/gas, and full frozen M4c variants on the same sample.

The best median absolute residual is obtained by the model `m4b_halo_star_gas`, with median absolute residual 0.0404 dex. The best fraction within 0.10 dex is obtained by `m4c_full_frozen`, with 83 of 87 galaxies within 0.10 dex. This indicates that the M4b variant gives the slightly sharper median model, while M4c gives the broader 0.10-dex containment.

## VIII. RESIDUAL STRUCTURE

The residual-correlation diagnostics are used as structure tests rather than as new tuning inputs. In the current results, earlier baselines show stronger residual structure tied

TABLE III. Baseline and ablation comparison on the same SPARC quality-1 sample.

| Model             | $N$ | Median abs. dex | Mean abs. dex | Max abs. dex | Frac. < 0.10 | Frac. < 0.15 | $N > 0.20$ | $N > 0.30$ |
|-------------------|-----|-----------------|---------------|--------------|--------------|--------------|------------|------------|
| newtonian_outer   | 87  | 0.249744        | 0.244856      | 0.448825     | 0.0689655    | 0.206897     | 57         | 31         |
| mond_simple_btfr  | 87  | 0.0472258       | 0.0561134     | 0.205226     | 0.83908      | 0.965517     | 1          | 0          |
| m3_scalar_q_only  | 87  | 0.0641311       | 0.0692546     | 0.211354     | 0.758621     | 0.896552     | 1          | 0          |
| m4a_halo_only     | 87  | 0.0416139       | 0.050371      | 0.23214      | 0.896552     | 0.977011     | 1          | 0          |
| m4b_halo_star_gas | 87  | 0.0404458       | 0.0464392     | 0.23214      | 0.931034     | 0.988506     | 1          | 0          |
| m4c_full_frozen   | 87  | 0.0416139       | 0.0457214     | 0.23214      | 0.954023     | 0.988506     | 1          | 0          |

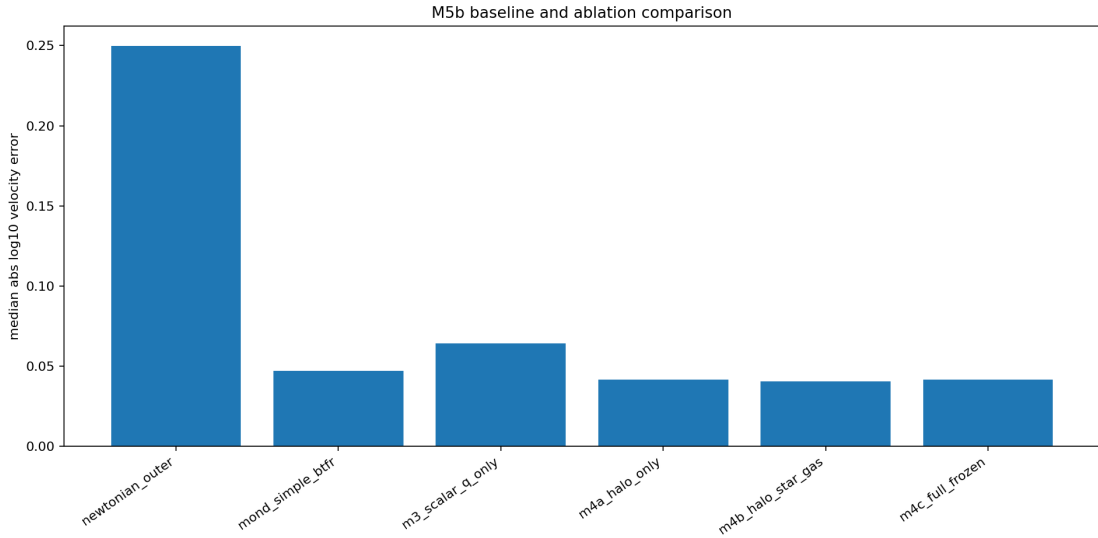


FIG. 2. Median absolute log-velocity residual for Newtonian, BTFR/MOND-like, and BDVR ablation models.

to stellar/gas composition, morphology, and velocity scale, while the full frozen response suppresses several of those trends.

## IX. RETAINED OUTLIER AND VERIFICATION TARGET

UGC07125 is retained in all metrics and is not tuned away. It is the leading residual object.

For UGC07125, the M5d inversion gives  $M_{\text{part,req}} = 8.8558e + 08M_{\odot}$ , compared with catalogued  $M_b = 7.5126e + 09M_{\odot}$  and  $M_* = 1.3560e + 09M_{\odot}$ . The required participating

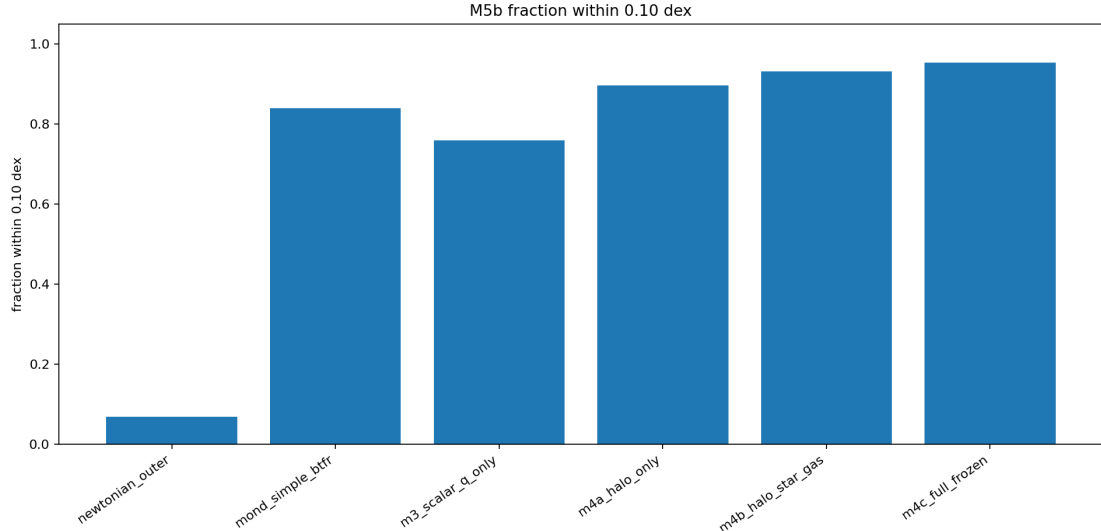


FIG. 3. Fraction of galaxies within 0.10 dex for baseline and BDVR ablation models.

fraction is 0.1179. This object is interpreted as a verification target rather than a resolved case.

The appropriate next checks are external: resolved HI kinematics, inclination and warp reliability, approaching/receding asymmetry, gas surface-density structure, and mass-estimate systematics.

## X. DISCUSSION

The current result supports BDVR as a nontrivial galaxy-sector effective law inside the Conformal Gravity Hilbert Space Time Current framework. The model is not a complete replacement for dark matter and is not a full rotation-curve fit. Its significance is that a frozen globally parameterized response law produces competitive flat-velocity residuals and meaningful retained failure modes.

The main theoretical task is to derive  $Q_0$ ,  $H$ ,  $S_{sg}$ ,  $S_{int}$ , and  $a_0$  from deeper bound-domain current variables rather than treating them as effective phenomenological factors.

## XI. CONCLUSIONS

We have defined BDVR as a galaxy-sector response law motivated by Conformal Gravity Hilbert Space Time Current bound-domain coupling. In frozen validation on a SPARC



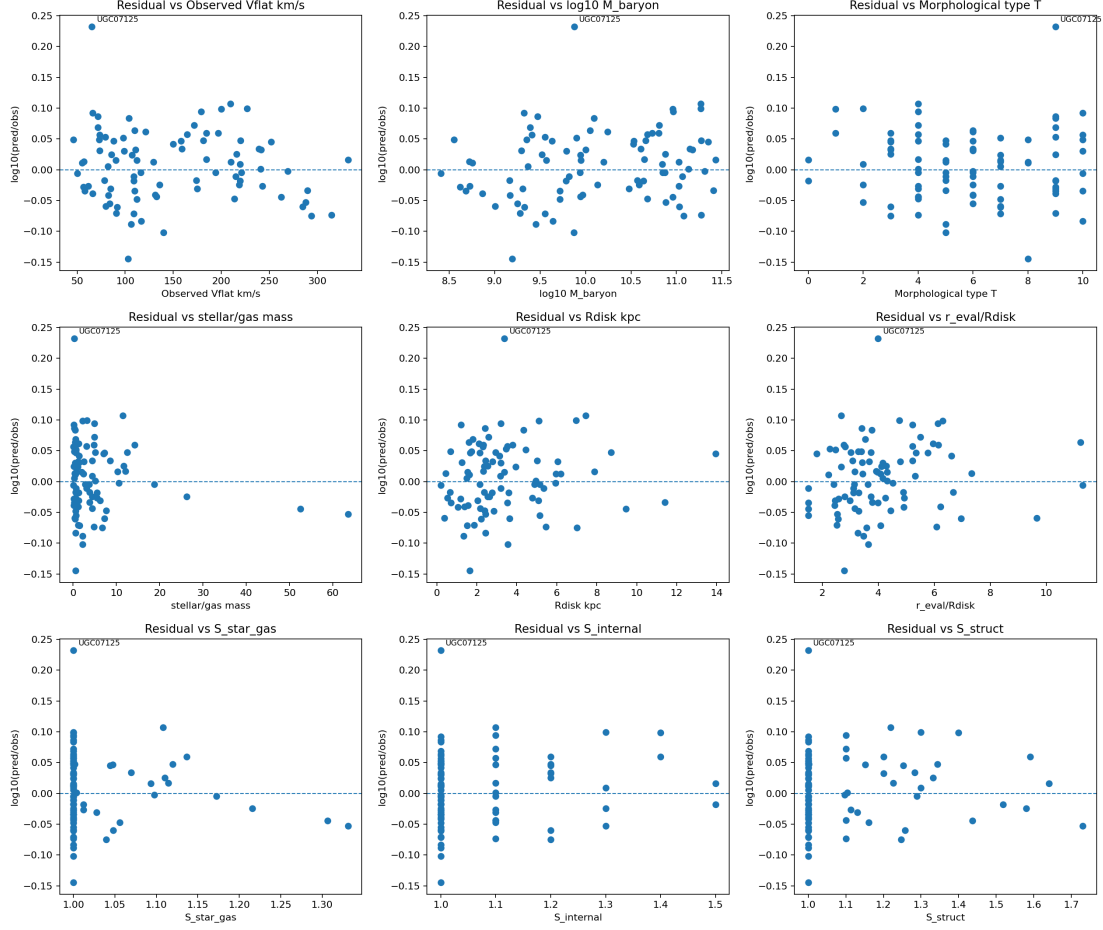


FIG. 4. Residual diagnostic dashboard for the frozen M4c validation sample.

quality-1 flat-velocity sample, the model achieves low residuals without per-galaxy  $Q$  tuning. Baseline and ablation comparisons show that radial halo activation and baryonic-structure conditioning are not merely decorative terms; they alter the residual behavior relative to simpler baselines. The leading residual object, UGC07125, is retained as a verification target and motivates external kinematic checks.

## Appendix A: UGC07125

## Appendix B: Reproducibility

The M6 artifact package records source file paths and hashes in `m6_source_artifacts_manifest.json`. The manuscript draft uses gathered outputs from M5 through M5d and introduces no new model logic.

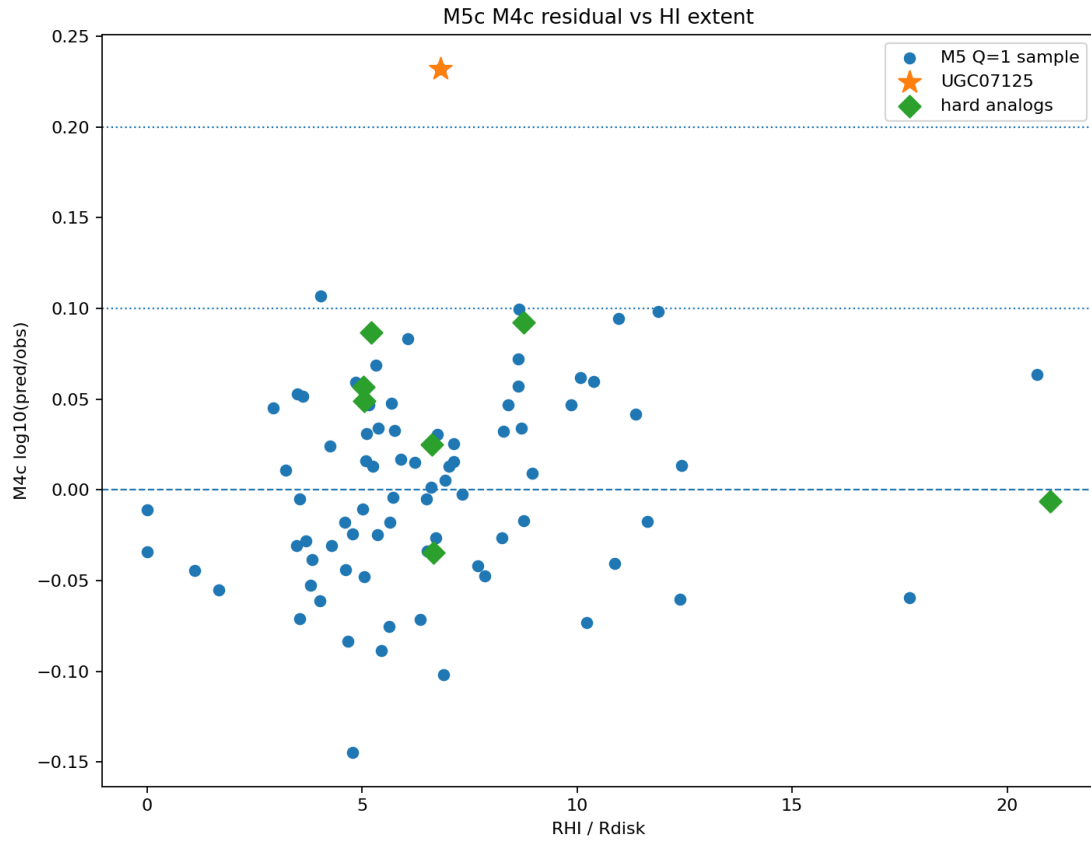


FIG. 5. UGC07125 analog-search residual behavior versus HI extent.

A final submission should include a data availability statement, code availability statement, run manifest, and archived reproducibility package.

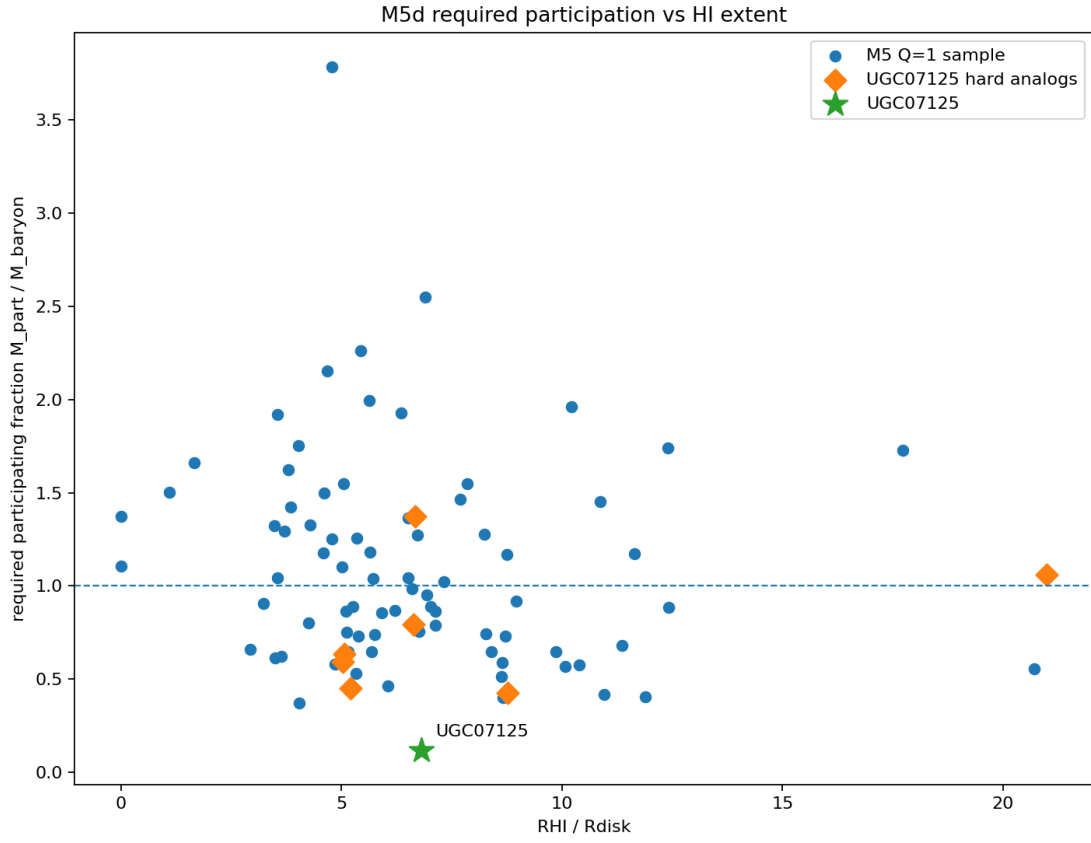


FIG. 6. Required participating mass fraction versus HI extent in the M5d inversion diagnostic.

TABLE IV. Largest residual correlations by model class. These diagnostics are used to characterize residual structure, not as additional tuning inputs.

| Model             | Variable             | $N$  | Pearson $r$ | Spearman $\rho$ | Slope       |
|-------------------|----------------------|------|-------------|-----------------|-------------|
| m3_scalar_q_only  | stellar_to_gas_ratio | 87   | -0.418072   | -0.377069       | -0.00275418 |
| m3_scalar_q_only  | star_frac            | 87   | -0.386556   | -0.377069       | -0.0872545  |
| m3_scalar_q_only  | gas_frac             | 87   | 0.386556    | 0.377069        | 0.0872545   |
| m3_scalar_q_only  | v_flat_obs_km_s      | 87   | -0.421025   | -0.376996       | -3.5407e-04 |
| m3_scalar_q_only  | S_star_gas           | 87   | -0.395256   | -0.373298       | -0.394874   |
| m4a_halo_only     | S_star_gas           | 87   | -0.424908   | -0.380006       | -0.433895   |
| m4a_halo_only     | stellar_to_gas_ratio | 87   | -0.444008   | -0.367427       | -0.0029898  |
| m4a_halo_only     | star_frac            | 87   | -0.383092   | -0.367427       | -0.0883871  |
| m4a_halo_only     | gas_frac             | 87   | 0.383092    | 0.367427        | 0.0883871   |
| m4a_halo_only     | v_flat_obs_km_s      | 87   | -0.412096   | -0.359244       | -3.5424e-04 |
| m4b_halo_star_gas | v_flat_obs_km_s      | 87   | -0.336134   | -0.305406       | -2.7025e-04 |
| m4b_halo_star_gas | star_frac            | 87   | -0.300895   | -0.283626       | -0.0649334  |
| m4b_halo_star_gas | gas_frac             | 87   | 0.300895    | 0.283626        | 0.0649334   |
| m4b_halo_star_gas | stellar_to_gas_ratio | 87   | -0.283907   | -0.283626       | -0.00178812 |
| m4b_halo_star_gas | S_star_gas           | 87   | -0.250682   | -0.264977       | -0.239431   |
| m4c_full_frozen   | M_gas_Msun           | 87   | 0.187678    | 0.330177        | 9.8263e-13  |
| m4c_full_frozen   | RHI                  | 87   | 0.226596    | 0.319846        | 8.7464e-04  |
| m4c_full_frozen   | H_halo               | 87   | 0.196081    | 0.202687        | 0.24409     |
| m4c_full_frozen   | r_over_Rdisk_eval    | 87   | 0.141987    | 0.202687        | 0.00445118  |
| m4c_full_frozen   | Reff                 | 87   | 0.117568    | 0.195483        | 0.0024987   |
| mond_simple_btfr  | stellar_to_gas_ratio | 87   | -0.418072   | -0.377069       | -0.00275418 |
| mond_simple_btfr  | gas_frac             | 87   | 0.386556    | 0.377069        | 0.0872545   |
| mond_simple_btfr  | star_frac            | 87   | -0.386556   | -0.377069       | -0.0872545  |
| mond_simple_btfr  | v_flat_obs_km_s      | 87   | -0.421025   | -0.376996       | -3.5407e-04 |
| mond_simple_btfr  | S_star_gas           | 87   | -0.395256   | -0.373298       | -0.394874   |
| newtonian_outer   | log10_M_baryon_Msun  | 87   | 0.637336    | 0.652092        | 0.0783836   |
| newtonian_outer   | M_baryon_Msun        | 87   | 0.505964    | 0.652092        | 7.8045e-13  |
| newtonian_outer   | star_frac            | 87   | 0.609233    | 0.650179        | 0.225421    |
| newtonian_outer   | gas_frac             | 1287 | -0.609233   | -0.650179       | -0.225421   |
| newtonian_outer   | stellar_to_gas_ratio | 87   | 0.394793    | 0.650179        | 0.00426331  |

TABLE V. UGC07125 retained-outlier audit fields and participating-mass diagnostic quantities.

| Field                             | Value                                                  | Source            |
|-----------------------------------|--------------------------------------------------------|-------------------|
| galaxy_name                       | UGC07125                                               | m5_prediction     |
| T                                 | 9.0                                                    | m5_prediction     |
| v_flat_obs_km_s                   | 65.2                                                   | m5_prediction     |
| v_m5_bdvr_km_s                    | 111.27246923349932                                     | m5_prediction     |
| velocity_ratio_pred_over_obs      | 1.7066329637039772                                     | m5_prediction     |
| log10_velocity_error              | 0.2321401297980161                                     | m5_prediction     |
| abs_log10_velocity_error          | 0.2321401297980161                                     | m5_prediction     |
| M_baryon_Msun                     | 7512569999.999999                                      | m5_prediction     |
| M_star_Msun                       | 1356000000.0                                           | m5_prediction     |
| M_gas_Msun                        | 6156569999.999999                                      | m5_prediction     |
| star_frac                         | 0.1804974862131068                                     | m5_prediction     |
| gas_frac                          | 0.8195025137868931                                     | m5_prediction     |
| stellar_to_gas_ratio              | 0.2202525107324371                                     | m5_prediction     |
| Rdisk                             | 3.38                                                   | m5_prediction     |
| RHI                               | 23.04                                                  | m5_prediction     |
| RHI_over_Rdisk                    | 6.816568047337278                                      | m5d_participation |
| Q_eff                             | 1.13195243693273                                       | m5_prediction     |
| H_halo                            | 1.2573763420608173                                     | m5_prediction     |
| S_star_gas                        | 1.0                                                    | m5_prediction     |
| S_internal                        | 1.0                                                    | m5_prediction     |
| S_struct                          | 1.0                                                    | m5_prediction     |
| M_part_required_Msun              | 885580433.570339                                       | m5d_participation |
| participation_fraction_required   | 0.11787982455675476                                    | m5d_participation |
| gas_participation_required        | -0.0764093588523579                                    | m5d_participation |
| gas_participation_required_status | stellar_mass_alone_exceeds_required_participating_mass | m5d_participation |